

Considering the investigator as a co-conversationalist in neurodivergent children’s use of discourse markers

Cynthia Boo^a, Nancy McIntyre^b, Peter Mundy^c, Letty Naigles^a

^a University of Connecticut, ^b University of Central Florida, ^c UC Davis, MIND Institute

Correspondence: cynthia.boo@uconn.edu



UConn Child Language Lab



UConn

UNIVERSITY OF CONNECTICUT

UC DAVIS

MIND INSTITUTE

BACKGROUND

- Discourse markers (DMs; e.g., *like*, *um*) can help speakers manage the flow and coherence of their verbal interactions with another person.^{1,2}
- Some research has reported that neurodivergent speakers, specifically individuals diagnosed with **autism** and/or **attention deficit/hyperactivity disorder (ADHD)** produce **fewer DMs** than their **neurotypical (NT)** peers, particularly when they engage in discourse with an adult investigator.³⁻⁵
 - These researchers argue that the paucity of DMs in neurodivergent individuals’ speech is related to their challenges with social communication.
- However, these studies did not consider **the role of the investigator as a co-conversationalist**.

OBJECTIVE: To investigate neurodivergent speakers’ use of DMs while simultaneously considering the role of the investigator as a co-conversationalist

METHODS

- Participants included:
 - 21 autistic children** ($M_{age}=11.6$, $SD=2.2$),
 - 24 ADHD children** ($M_{age}=11.9$, $SD=2.5$), and
 - 22 NT children** ($M_{age}=12.5$, $SD=2.3$).
- Language samples were collected using a virtual reality paradigm where children viewed a classroom as they answered questions about themselves (e.g., “What is a normal day like for you?”).
- Audio recordings were transcribed via CLAN.⁶
- DMs were identified via utterance-by-utterance coding following Schiffrin’s (1987) approach.
- A total of **13 DM types** (*actually*, *and*, *because*, *but*, *I mean*, *like*, *now*, *so*, *then*, *uh*, *um*, *well*, & *you know*) and **over 4,500 DM tokens** (both child and investigator) were identified in the entire corpus.

RESULTS

Fig. 1. DM Tokens:⁺
Autistic & ADHD < NT

$F(2, 64) = 5.047$, $p = 0.009$, $\eta^2 = 0.136$

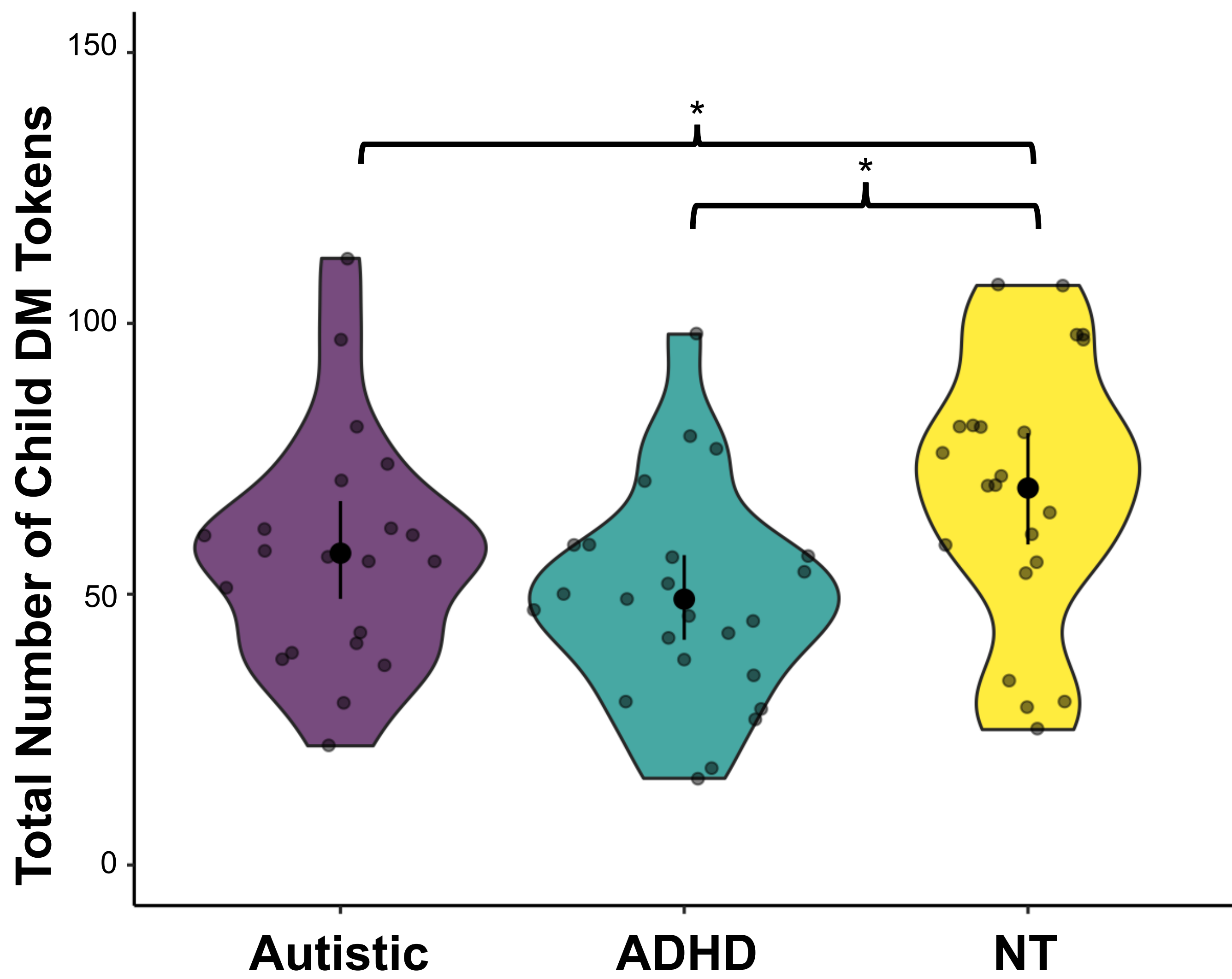


Fig. 2. POSITIVE relationship between investigator and autistic children’s use of DMs⁺

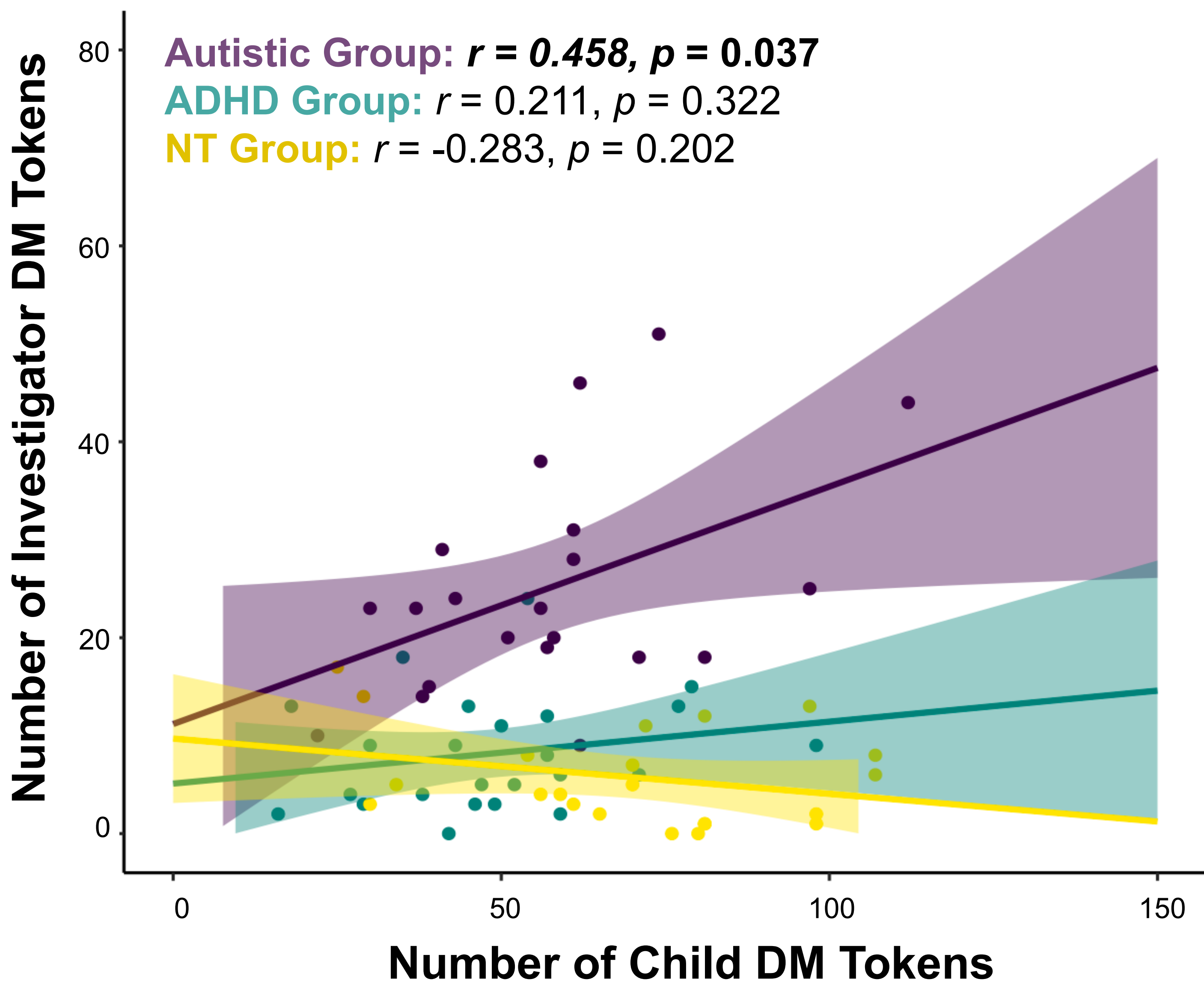
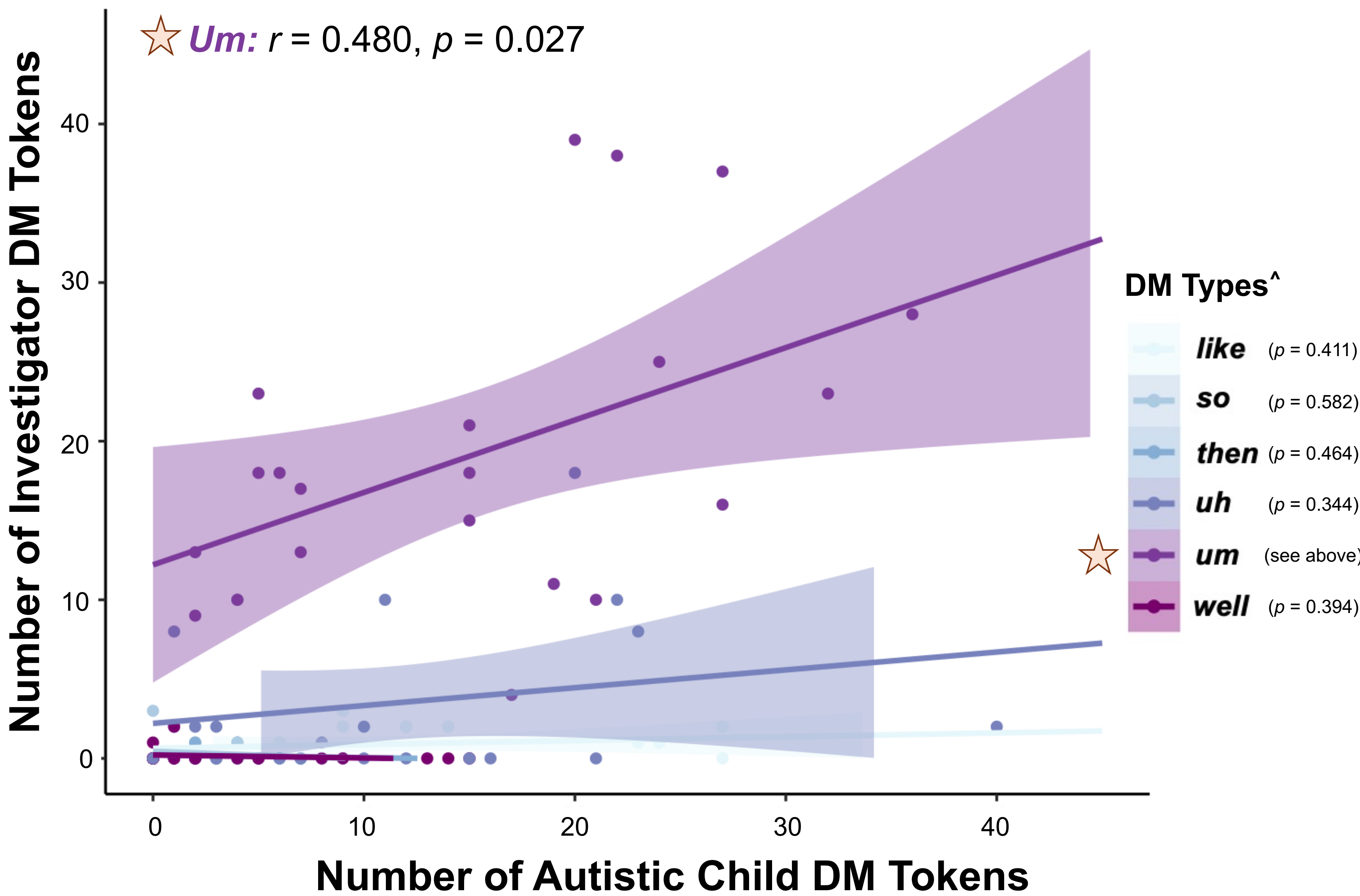


Fig. 3. POSITIVE relationship between investigator and autistic children’s use of *um*



⁺ additional analyses were conducted using proportion of DM tokens per utterance; with the same patterns as reported above, we have chosen to present the raw number of tokens

[^] no investigator used the other DM types (*actually*, *because*, *but*, *I mean*, *now*, or *you know*); thus, post hoc analyses focused on the six DM types above

DISCUSSION

- Autistic children’s use of DMs, specifically “um,” was systematically related to that of the investigator.**
 - This could be the result of **possible reciprocal priming effects**.
 - While the investigator followed a script in this study, **the use of DMs is often unscripted and naturally-occurring** – more DMs might be needed to manage the flow and coherence of verbal interactions between investigator and autistic children.
 - Additionally, this finding hints at **possible pragmatic strengths** of autistic speakers, as they are attuned to their co-conversationalist’s speech.

REFERENCES

- Fox Tree & Clark, 2002. *Cognition*.
- Schiffrin, 1987. *Discourse markers*.
- Irvine et al., 2016. *JADD*.
- Lawley et al., 2022. *JADD*.
- Redmond, 2005. *Clinical Linguistics & Phonetics*.
- MacWhinney, 2000. *The CHILDES Project: Tools for analyzing talk*.

ACKNOWLEDGEMENTS

We thank the children and their families who participated in this research, as well as the dedicated students who transcribed the audio files. This research was supported by grants from the **National Institute on Deafness and Other Communication Disorders (R01DC016665 to L. Naigles)** and the **Institute of Education Sciences (R324A110174 to P. Mundy)**.